Neural Networks and Deep Learning

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What is a Neural Network?

A neural network consists of layers of interconnected nodes called neurons. The Input Layer receives raw features. Hidden Layers learn intermediate representations through nonlinear transformations. The Output Layer produces the final prediction. Each connection has a weight W and each neuron has a bias b. Output = activation(W*x + b).

Activation Functions

ReLU (Rectified Linear Unit): f(x) = max(0, x). Most common in hidden layers, computationally efficient, mitigates vanishing gradient. Sigmoid: f(x) = 1/(1+exp(-x)), outputs [0,1], used for binary output. Softmax: converts logits to class probabilities summing to 1, used for multi-class output. Tanh: outputs [-1,1], used in recurrent networks.

Backpropagation and Optimizers

How neural networks learn: (1) Forward Pass - compute prediction. (2) Loss function measures error: Binary Cross-Entropy for binary classification, Categorical Cross-Entropy for multi-class, MSE for regression. (3) Backward Pass - compute gradients via chain rule. (4) Update weights using optimizer. SGD: w = w - lr * gradient. Adam: adaptive learning rates per parameter, most widely used optimizer.

Common Architectures

Feedforward Neural Networks (FNN): Data flows input-to-output. Good for tabular data. CNN (Convolutional Neural Networks): Convolutional layers detect spatial features (edges, shapes, textures) in images. Used for image classification and object detection. RNN/LSTM/GRU: Process sequential data, handle time series and text. LSTM gates control information flow, solving vanishing gradient. Transformers: Self-attention mechanism, powers BERT, GPT, LLaMA, and modern GenAI.

Regularization Techniques

Dropout: Randomly set a fraction of neuron outputs to 0 during training, preventing co-adaptation. Batch Normalization: Normalize inputs of each layer, stabilizes and accelerates training. L2 Regularization (Weight Decay): Add penalty proportional to squared weights to loss function. Early Stopping: Monitor validation loss and stop training when it stops decreasing to prevent overfitting.

Key Hyperparameters

Learning Rate: Controls the step size for weight updates. Too high causes divergence, too low causes very slow convergence. Common values: 0.001-0.0001 with Adam. Batch Size: Number of samples per weight update. Larger batches are faster but may generalize worse. Epochs: Full passes through the training data. Number of layers and neurons determines model capacity.